

Scanned File

*Ben Rich
Lockhead*

Subj: Fw [REDACTED] stuff from [REDACTED]
Date: 05/19/2000 12:21:21 AM Eastern Daylight Time
From: [REDACTED]
To: [REDACTED] (Steven M. Greer)

*Wrote
to Rich
Talk
at UCLA*

THE SKUNK WORKS

We have some new things. We are not stagnating. What we are doing is updating ourselves, without advertising. There are some new programs, and there are certain things — some of them 20 or 30 years old — that are still break-throughs and appropriate to keep quiet about. Other people don't have them yet. [1]

Ben Rich

The Antelope Valley is about ___ miles north of Los Angeles on Highway 14. Going north just after Palmdale you can see an enormous box of a building on the right. Just a few feet east of the Sierra Highway is Lockheed-Martin's Advanced Development Company. Most local people know it simply as the "Skunk Works." On the side facing the freeway is a small, characterized, cartoon of a black and white skunk. I ought to know, I lived in the Antelope Valley for six years. The Skunks Works got its name from Al Capp's Little Abner nationally-syndicated comic strip. Capp's comic strip featured a backwoods, hillbilly still, called "the skonk works", which fermented "kickapoo joy juice" from dead skunks and old leather shoes. Back in the early 1940's Lockheed was located in Burbank, California and was headed up by an engineer named Clarence "Kelly" Johnson. The war was on and Johnson's small, elite engineering group was housed in a rented circus tent adjacent to a odorous plastics factory. One day, the telephone rang and someone answered "skonk works." They have been called the "Skunk Works" ever since. The skunk cartoon is even painted on some of their planes.

Kelly Johnson's group is probably the most famous aircraft design "studio" ever known. For those who are not aware of their story, the Skunk Works produced the P-38 Lightning fighter, called the "Forked-Winged Devil" by German pilots. It was used quite effectively to shoot down Admiral Yamamoto's personal transport plane helping to bring the Pacific War to a somewhat earlier climax. Johnson's group also produced the P-80 Shooting Star, America's first operational jet fighter and the F-104 Starfighter, the first supersonic jet attack plane. But Johnson's work will be remembered by most for the creation of the Lockheed U-2 spy plane and the one plane that almost everyone has heard of, the SR-71 Blackbird, both reconnaissance aircraft developed in the fifty's and sixty's. The Skunk Works seems to have had a lot of "firsts". The Skunk Works is not a think tank. What the Skunk Works excels at is designing and fabricating America's cutting-edge, aerospace craft. Today, it's working on the X-33 Venture Star, the first single-stage-to-orbit spacecraft — the future replacement of the Space Shuttle. At least, that is what it is willing to admit to in public. When Kelly Johnson retired in 1974, his hand-picked successor, Ben Rich, was selected to sit in the pilot's seat. A wise choice.

To get a better feeling of Rich, the man, and Rich, the craftsman, it's best

to hear it in his own words. In 1994 [REDACTED] with [REDACTED] authored A Personal Memoir of My Years at Lockheed. [2] These are [REDACTED] words:

[REDACTED] actually began in July, 1975, about six months after I took over the Skunk Works. I attended one of the periodic secret Pentagon briefings held to update those with a need to know on the latest Soviet technical advances in weapons and electronics.

"The United States had only two defensive ground-to-air missile systems — the Patriot and the Hawk, both only so-so in comparison to the Soviet weapons. By contrast, the Russians deployed 16 different missile systems to defend their cities and vital strategic interests. Their early-warning radar systems, with 200-foot-long antennas, could pick up an intruding aircraft from hundreds of miles away. Their SAM ground-to-air missile batteries were able to engage both low-flying attack fighters and cruise missiles at the same time. The Soviet SAM-5, a defensive surface-to-air missile of tremendous thrust, could reach heights of 125,000 feet and could be tipped with small nuclear warheads. So these weapons systems posed a damned serious threat.

"Most troublesome, the Russians were exporting their advanced nonnuclear defensive systems around the world. The Syrians now had nonnuclear SAM-5s. And during our Pentagon briefing we heard a chilling analysis of the 1973 Yom Kippur war involving Israel, Syria, and Egypt. Although the Israelis flew our latest and most advanced jet attack aircraft and their combat pilots were equal to our own, they suffered tremendous losses against an estimated arsenal of 30,000 Soviet-supplied missiles. The Israelis lost 109 airplanes in 18 days, mostly to radar-guided ground-to-air missiles and antiaircraft batteries, manned by undertrained and often undisciplined Egyptian and Syrian personnel.

"What really rattled our Air Force planners was that the evasive maneuvering by Israeli pilots to avoid missiles — the same tactics used by our own pilots — proved to be a disaster. All the turning and twisting calculated to slow down an incoming missile made the Israeli aircraft vulnerable to conventional ground fire. If the Israeli loss ratio were extrapolated into a war between the United States and the highly trained Soviet Union and Warsaw Pact nations in Eastern Europe, a war fought using similar airplanes, pilot training, and ground defenses, our air force could expect to be decimated in only 17 days.

"I was not too surprised. The Skunk Works had firsthand experience with the latest Soviet equipment because the CIA had scored spectacular covert successes in acquiring Soviet hardware by one means or another. We could not only test their latest fighters or new radars and missile systems but actually fly against them. Skunk Works technicians pulled these systems apart, then put them back together, so we had a sound notion of what we were up against.

"That Pentagon briefing was particularly sobering because it was one of those rare times when our side admitted to a potentially serious gap that tipped the balances against us. I had our advanced planning people noodling all kinds of fantasies — pilotless, remote controlled drone tactical bombers and hypersonic aircraft that would blister past Soviet defenses at better than five times the speed of sound, once we solved awesomely difficult technologies.

"I wish I could claim to have had a sudden 2 a.m. revelation that made me bolt upright in bed and shout, 'Eureka!' But the truth is that an exceptional 36-year-old Skunk Works mathematician and radar specialist named [REDACTED] decided to drop by my office one April afternoon and presented me with the Rosetta Stone breakthrough for stealth technology. "The gift he handed to me over a cup of decaf instant coffee would make an attack airplane so difficult to detect that it would be invulnerable against the most advanced radar systems yet invented, and survivable even against the most heavily defended targets in the world.

"Denys had discovered this nugget deep inside a long, dense technical paper on radar written by one of Russia's leading experts and published in Moscow nine years earlier. The author was [REDACTED] chief scientist at the Moscow Institute of Radio Engineering. As Denys admitted, the paper was so obtuse and impenetrable, only a nerd's nerd would have waded through it all, underlining yet!

"[REDACTED] saw my blank stare. Radar cross section calculations were a branch of medieval alchemy as far as the noninitiated were concerned. Making big objects appear tiny on a radar screen was probably the most complicated, frustrating, and difficult part of modern warplane designing. A radar beam is an electromagnetic field, and the amount of energy reflected back from the target determines its visibility on radar. For example, our B-52, the mainstay long-range bomber of the Strategic Air Command for more than a generation, was the equivalent of a flying dairy barn when viewed from the side on radar.

"[REDACTED] has shown us how to create computer software to accurately calculate the radar cross section of a given configuration, as long as it's in two dimensions," Denys told me. "We can break down an airplane into thousands of flat triangular shapes, add up their individual radar signatures, and get a precise total of the radar cross section."

"Why only two dimensions, and why only flat plates? Simply because, as Denys later noted, it was still only 1975 and computers weren't yet sufficiently powerful in storage and memory to allow for three-dimensional designs, or rounded shapes, which demanded enormous numbers of additional calculations.

"The result we called 'faceting' — creating a three-dimensional airplane design out of a collection of flat sheets or panels. The Skunk Works would be the first to try to design an airplane composed entirely of flat, angular surfaces. I tried not to anticipate what some of our crusty old aerodynamicists might say. What emerged was a diamond beveled in four directions, creating in essence four triangles. Viewed from above, the design closely resembled an Indian arrowhead.

"'Boss', Denys said, handing me the diamond-shaped sketch. 'Meet the Hopeless Diamond.'

"'If we made this shape into a full-size tactical fighter, what would be its equivalent radar signature? As big as what — a Piper Cub, a T-38 trainer — what?'

"Denys shook his head vigorously. 'Ben, understand, we are talking about a major, major, big-time revolution here. We are talking infinitesimal.'

"Well,' I persisted, 'What does this mean? On a radar screen it would appear to be as big as a condor, an eagle, maybe an owl, a what?'

"Ben,' he replied with a loud guffaw, 'try as big as an eagle's eyeball.'

"Kelly Johnson was not impressed. Unfortunately, he caught me leaning over a worktable studying a blueprint, and I never heard him coming. Kelly kicked me in the butt - hard too. Then he crumpled up the stealth proposal and threw it at my feet. 'Ben Rich, you dumb shit,' he stormed. 'Have you lost your mind?' This crap will never get off the ground.'

"At the Skunk Works we designed practical, used off-the-shelf parts whenever possible, and did things right the first time. My wind man, for example, had designed twenty-seven winds on previous Skunk Works' airplanes before tackling the Hopeless Diamond. All of us had been trained by Kelly Johnson and believed fanatically in his insistence that an airplane that looked beautiful would fly the same way. No one would dare to claim that the Hopeless Diamond would be a beautiful airplane. As a flying machine it looked alien.

"Dick Cantrell, head of our aerodynamics group, suggested burning Denys at the stake as a heretic. But after a couple of hours of listening to Overholser's explanations of stealth, Dick dropped his lanky frame onto the chair across from my desk and heaved a big sigh. 'OK, Ben', he muttered. 'I surrender. If that flat plate concept is really as revolutionary as that kid claims in terms of radar cross section, I don't care what it looks like. I'll get that ugly son-of-a-bitch to fly.'

"We could get the Statue of Liberty to do barrel rolls with the onboard computers that achieved aerodynamic capability by executing thousands of tiny electrohydraulic adjustments every second to an airplane's control surfaces. But even with the powerful onboard computers, getting into the sky, as Kelly's boot to my butt suggested, would be far from a cakewalk.

"Several of our aerodynamics experts, including Dick Cantrell, seriously thought that maybe we would do better trying to build an actual flying saucer. The shape itself was the ultimate in low observability. The problem was finding ways to make a saucer fly. Unlike our plates, it would have to be rotated and spun. But how? The Martians wouldn't tell us.

"We had a ten-foot wooden model of the diamond, and we took it and the original wooden model for the manta-ray-shaped D-21 drone and put them side by side into an electromagnetic chamber and cranked up the juice. Hopeless Diamond was exactly as Denys had predicted: a thousand times stealthier than the 12-year-old drone. The fact that the test results matched Denys' computer calculations was the first proof that we actually knew what in hell we were doing.

"Our next big hurdle was to test the model on an outdoor radar test range near Palmdale, on the Mojave desert. Our model was mounted on a 12-foot-high pole, and the radar dish was zeroed in from about 1,500 feet away. I was standing next to the radar operator in the control room. 'Mr. Rich, please check on your model. It must've fallen off the pole,' he said.

"I looked. 'You're nuts,' I replied. 'The model is up there.' Just then a blackbird landed right on top of the Hopeless Diamond. The radar operator smiled and nodded. 'Right, I've got it now.' I wasn't about to tell him he

was zapping a crow. His radar wasn't picking up our radar at all. For the first time, I felt reassured that we had caught the perfect wave at the crest and were in for one terrifically exciting ride. I saw firsthand how invisible that diamond shape really was.

"It's the first of December 1977, just after sunup, the best time for test pilots to take off. Winds are usually calmest then, but this morning the chill wind blasts through my topcoat like it's tissue paper. This flight will be every bit as important to the nation's future and the future of the Skunk Works as the first test flight of the U-2 spy plane, which took place at this very same remote sand pile more than a quarter century ago.

"Back then, I was a Skunk Works rookie and this base, which we built for the CIA, was just a tiny outpost of windswept quonset huts and trailers, guarded by CIA agents with tommy guns. Kelly had jokingly nicknamed this godforsaken place Paradise Ranch, hoping to lure young and innocent flight crews to work on a dry lake bed where quarter-inch rocks blew around most afternoons. It is now a sprawling facility, bigger than some municipal airports, a test range for sensitive aviation projects. No one nowadays gains access without special clearances that include a polygraph test. Such paranoia has kept our most guarded national defense secrets secret. [3]

"Today, the Have Blue prototype that will soon be rolling down this runway is the first built under my regime after Johnson's retirement three years earlier. But we really aren't one hundred percent certain that this sucker can actually get off the ground. It is the most unstable and weirdest-looking airplane since Northrop's flying wing, built on a whim back in the late 1940's.

"Bill Park, our chief test pilot, complained that it was the ugliest airplane he'd ever strapped himself into. To Bill, even the opaque triangular cockpit is ominous, especially if he has to punch out. But the specially coated glass will keep radar beams from picking up his helmeted head. The real beauty of Have Blue is that Bill's head is a hundred times more observable on radar than the airplane he will be flying.

"Kelly Johnson is standing at my side, looking stoic. He's still skeptical about whether or not this prototype will prove way too draggy to get off the ground.

"Bill pushes on the throttle, and Have Blue slowly begins to accelerate. To stay stealthy, Have Blue has no afterburners, and it will need almost as much runway as a 727 loaded with fuel, baggage, and passengers bound for Chicago. Bill goes full throttle. He's chewing up a lot of runway as he sweeps past us.

"Uh-oh. He's damn near off the end of the runway. Then I see him lift off. Slow as a jumbo jet a hundred times its weight, but he's up. Get up. Up, up, up. The little airplane hears me. It's heading toward the snow-powdered mountains.

"Kelly slaps me on the back and shouts, 'Well, Ben, you got your first airplane.'

"Not so fast. There are some significant foothills looming in  light path and I try to do some quick mental calculations to get him safely over the hump. I raise my binoculars and quickly try to adjust the focus. By the time the mountains come clear, our airplane is across the other side.

"Stealth was our great good fortune, and our earnings skyrocketed. The stealth fighter brought in more than \$6 billion. By my fifth year, I was heading a small, secret R&D outfit whose annual earnings placed it among the Fortune 500. Not bad. Not bad at all."

The rest is history. During the Gulf War, Lockheed's F-117A Nighthawk stealth attack planes, derived from the Have Blue prototype, returned from Baghdad without a hit or even a near miss. [REDACTED] stealth program worked.

One day back in 1993, I received a call from a good friend, [REDACTED] had graduated in engineering at UCLA back in the seventies (about a decade after me). [REDACTED] said, "How'd you like to hear a lecture from someone from Lockheed's Skunk Works?" I asked, "Sure, who's speaking?" "[REDACTED]" "Who's [REDACTED]", I asked, never having even heard his name.

On the appointed day (March 23, 1993), I parked my car in the parking structure on Gayley Avenue and walked over to the UCLA faculty center. There was a small crowd gathering outside the closed doors to the auditorium. I met [REDACTED] as he walked through the door at the appointed time. I asked him how he got on the invitation list. He was still active in the UCLA Engineering Alumni Association and this was one of the events that they had planned. We sat at the front of the audience, say about 150 to 200 folks. I took a look around. The 250 or so capacity auditorium was filled by what appeared to be aerospace industry professionals, the press, members of academia and a few military types in uniform sitting at the back. Air Force blue-suiters. Knowing the history of Lockheed's activity in military intelligence, my guess was that there were also a few "spooks" there too. No one had an ID badge saying "chief spook" for this agency or that, but it just figured. This was soon to be confirmed.

I had no idea what [REDACTED] looked like or who exactly he was. Of course, I knew of Kelly Johnson, but I also knew he had retired. Soon, a short, 5' 5" or so, thin, gray-haired, gentlemen walked up to the podium. He was a wiry-looking man with sixty-five years of worry wrinkles on his face and wearing gray slacks and a white shirt. It turned out, at least for me, that he seemed like the kind of boss that you wanted to keep on your side. Before he said anything more, his first words were, "Please don't ask me any questions about the Aurora project. [4] I can't answer any and if I did, I'd be thrown in jail. There are some representatives of the CIA here who I recognize."

Contrary to Kelly Johnson, who had little to no humor, [REDACTED] confesses to being a bit of a jokester. In his book, Rich acknowledges that he had "an endless supply of one-liners and farmer's daughters jokes supplied fresh daily by my brother, a television producer on a situation comedy." But was [REDACTED] kidding about loose lips and jail time? At the time, Rich sounded dead serious to me. My feeling is that Kelly Johnson had instilled in him almost a fear and certainly a serious respect for national security. Again, from his book he quoted Kelly Johnson as saying, "[REDACTED] this project [the U-2 spy plane] is so secret that you may have a six-month to one-year hole in your resume that can never be filled in. Whatever you learn, see, and hear for as long as you work inside this building stays forever inside this building. Is that clear? You'll tell no one about what we do or what you do — not your wife, your mother, your brother, your girlfriend, your

priest, or your CPA. You got that straight?'

"Yes, sure, [redacted] relied.

"Okay, first read over this briefing disclosure form which says what I've just said, only in governmentese. Just remember, having a big mouth will cost you twenty years in Leavenworth, minimum." Was Rich just joking? I don't think so. [2]

[redacted] went on for an hour reviewing the history of the Skunk Works. It was obvious from the way he carried himself and what he said that he was proud of his accomplishments. His lecture was highlighted by numerous slides of each project that he had worked on. He described the U-2 reconnaissance plane and its successor, the TR-1 variants. He spoke about the SR-71 Blackbird, its various missions, its speed records and a few of its less well-known mishaps. He talked about the little-known, D-21 supersonic spy drone that was carried atop an SR-71 mother ship. He even spoke of missions in which the D-21 was launched toward mainland China and went out of control. He covered the Sea Shadow, a prototype stealth surface ship commissioned by the Navy (according to him, the Navy ran it through a few sea trials, but never really understood how to use it). The part of the lecture in which he truly beamed was the F-117 stealth attack plane (sometimes erroneously called a stealth "fighter" which it is not. It's strictly an attack bomber and is subsonic.) All of these aircraft were built by the Skunk Works, not designed by them and then built by someone else. The Skunk Works were R&D designers but they were also fabricators. In other words, they were also "nuts and bolts" people. Here it is in Rich's own words: "Inside the Skunk Works, we were a small, intensely cohesive group consisting of about fifty veteran engineers and designers and a hundred or so expert machinists and shop workers. Our forte was building a small number of very technologically advanced airplanes for highly secret missions." [5]

Occasionally, [redacted] would say a few surprising words. At one point of his lecture he said, "If you can imagine it, Lockheed Skunk Works has done it." At the time this seemed a little odd to me. Why would a person of [redacted] stature say that? It seemed that it was either a brag (which really wasn't necessary) or he was trying to say something euphemistically. In fact, listening to [redacted] I got the distinct impression that there was so much more that he wanted to announce to the world, but, of course, couldn't. [redacted] repeated the "If you can imagine it." phrase twice more during the lecture.

During the hour or so talk, probably the most fascinating lecture that I have ever witnessed [redacted] had shown a carousel-full of slides. He made a few concluding comments. Then he said that the Skunk Works would continue to work on cutting-edge technology into the future. He said no one knows what we may work on next (or something to that effect). He then showed his last slide. The last slide was an artist's conception of a flying disk-shaped craft — a "flying saucer." At this point, [redacted] said, "We now have the technology to take ET home." The reaction of the audience was nervous laughter. My interpretation was that this was Rich's way of saying, "We can do it now." A few others in the room, including my friend Jan and myself, took it very seriously. All you had to do was read between the lines.

The talk concluded. About 30 of us crowded around [redacted] like bobby-soxers around The Beatles at their last concert. I was beginning to wonder if any

one else (except [redacted] as he and I think alike) in the entire audience was having the same thoughts as I. Could the Skunk Works be working on some otherworldly craft as we spoke? One man spoke up and said that he was a new manager at Northrop. [redacted] pointed his finger at the man and said, "Well let me ask you a question: Is it possible to travel to the stars?" The man was taken back a bit and then said, "Oh sure, it would just take a long time."

[redacted] answer to that was "No, it won't take someone's lifetime to do it."

"There is an error in the equations (Jan and I had different reactions: [redacted] was thinking Maxwell or Einstein and I assumed he was speaking strictly Einstein) and we know what it is, and now have the capability to travel to the stars." A man in the crowd asked, 'Just how are we going to get to the stars?' [redacted] immediately responded, "First, you have to understand that we will not get to the stars using chemical propulsion." In other words, no rocket ships. "Second, we have to devise a new propulsion technology. What we have to do is to find out where Einstein went wrong." [redacted] went on to imply that various people at the Skunk Works had been studying alternative propulsion technologies for interstellar travel. Rich said, they had, for example, determined that Einstein's equations dealing with relativity theory were incorrect. I asked him to clarify that. Did he mean that the Skunk Works employed theoretical physicists, "Einstein types", to look for alternate means of space travel? [redacted] said, "Yes." He went on to say that they had proved that Einstein was wrong. He made a mistake. I didn't know how that set with other people in the room, but to me that possibly meant that they had determined how it was possible to travel faster than the speed of light. At this point, one must keep in mind one thing. The Skunks Works is a fabrication outfit. They are not in business to be a theoretical think tank. My take on this was that they were looking for loopholes in Einstein's work that would give them a way of traveling at ultra-high speeds, faster-than-light travel, to the stars and back. I took it to mean that they had found it. And they were building or had built a craft to do it. One woman in the small crowd around him apparently had similar thoughts. She asked, "Mr. [redacted] when will the ETs go home?" [redacted] smiled, looked down at the floor and said nothing. Some of us were left wondering what his response might have been had he been able to answer without being sent to do twenty years in Leavenworth.

As Rich was walking out the door, [redacted] followed him and asked privately, "Ben, what equations are you talking about?" He just looked at [redacted] [redacted] explained that he was interested in propulsion and how it worked. [redacted] then asked him, "How does ESP work?" [redacted] replied without thinking, "All points in time and space are connected." [redacted] shot back, "That's how it works!" [redacted] did not know if he was referring to his question or his answer when he said "That's how it works." [redacted] then turned around and exited abruptly. [The answer to this issue is in Chapter XX]

Rich died of cancer in 1996. Who knows what secrets he took to the grave?

NOTES

1. Brown, Stuart F. and James C. Goodall, "The Supersonic Spy Drone", Popular Science, October 1994.
2. Rich, Ben R and Leo Janos, Skunk Works: A Personal Memoir of My Years at Lockheed, Little Brown and Co., 1994 and "Inside the Skunk Works", Popular Science, October 1994.
3. For the uninitiated or for those who have not as yet guessed, the mysterious desert base that Rich is referring to, his "Paradise Ranch", is located in Nevada. It's called "Area 51."

5. See note 2.

XXXXXXXXXXXX0

[illegible]